

Daniel Foister, Jason Gun, Timyathus Spikes, Tyrese Dickson - DTJT Group
Shehenaz Shaik
CSCI 4350
Sprint #4 - 2/26-3/5

Roles:

- Product Owner: Tyrese Dickson
- Scrum Master: Jason Gun
- Development Team: Daniel Foister, Timyathus Spikes

Time Limit: Two Days (2/26, 3/3, 3/5)

Attendance: All group members were present this week.

Project Github Repo: <https://github.com/12061062/StockBuddy>

Sprint Goal:

- The goal for this sprint was to improve the usability, as best we could, of the POS system by implementing product search features, barcode scanning integration, UI consistency improvements, and low-stock alert notifications. These improvements help to improve our checkout operations and inventory awareness for store managers.

PBIs/User Stories/To-Do Chosen for Sprint:

- **SCRUM-31:** Ensure the frontend UI is consistent between each page and follows our style guidelines.
 - Estimation: 4 story points
- **SCRUM-32:** Search for products via name or SKU if the barcode scanner fails.
 - Estimation: 5 story points
- **SCRUM-33:** Scan barcodes with a barcode scanner, and have the item automatically added to the cart.
 - Estimation: 5 story points
- **SCRUM-34:** Low stock level alerts.
 - Estimation: 3 story points
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- The PBIs/tasks selected for this sprint were based on the team's average velocity of around "15 story points" per sprint, based on previous sprints. These features were prioritized because they directly improve the usability of our POS app. Barcode scanning and manual searches ensure checkout can continue even if the hardware fails, while the UI consistency and stock alerts improve our inventory awareness and how our product looks.

Sprint Planning

- During sprint planning, our team reviewed the backlog and selected/created four PBIs/tasks that aligned well with StockBuddy's current development stage. The tasks were distributed evenly among our team members, and the acceptance criteria were clarified before beginning implementation.

Total Sprint Duration:

- 4 members x 8 hours/week x 1 Week/Sprint = **32 man-hours available**

Tasks Performed:

- Task Name: Ensure the frontend UI is consistent between each page and follows our style guidelines.

- Task Owner: Tyrese Dickson
- Planned Duration: 4 hours
- Actual Duration: 5 hours
- Status: Completed
- Challenges: None
- Lessons Learned: Maintaining a consistent UI early prevents a lot of rework/headache later and improves the overall usability.

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- Task Name: Search for products via name or SKU if the barcode scanner fails.
 - Task Owner: Timyathus Spikes
 - Planned Duration: 5 hours
 - Actual Duration: 6 hours
 - Status: Completed
 - Challenges: Making sure the search functionality returned accurate results quickly took me longer than expected. I had to refine the database queries.
 - Lessons Learned: Implementing fallback features such as manual search improves the overall system.

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- Task Name: Scan barcodes with a barcode scanner, and have the item automatically added to the cart.
 - Task Owner: Jason Gun
 - Planned Duration: 5 hours
 - Actual Duration: 6 hours
 - Status: Completed
 - Challenges: Had to complete additional testing to make sure the items were completely identified with the barcode input.
 - Lessons Learned: None that I identified.

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- Task Name: Low stock level alerts.
 - Task Owner: Daniel Foister
 - Planned Duration: 5 hours
 - Actual Duration: 6 hours
 - Status: Completed
 - Challenges: None
 - Lessons Learned: Alert systems need to balance both usefulness and 'noise' on the screen to remain helpful.
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Sprint Review:

- Was the Sprint Goal Achieved?:
 - The Sprint Goal was Achieved.
 - Our DoD: Our work/code maps to all requirements specified in the Jira Board (backlog).
 - Our team successfully implemented UI consistency improvements, product search functionality, barcode scanning integration, and low-stock alerts. These improvements really help our checkout efficiency and

provide better inventory awareness for store managers and checkout clerks.

- Focus for the Next Sprint:
 - Begin implementing reporting features for sales and inventory.
 - Improve UI feedback for alerts and product selection.
 - Expand/add more automated tests.
- PO/Stakeholder Review:
 - The Product Owner reviewed all completed tasks/backlog items:
 - The frontend UI is consistent across the system pages and follows the team's style guidelines.
 - Barcode scanning correctly adds items directly to the cart during checkout.
 - Products can be searched by name or SKU when barcode scanning is unavailable.
 - Low stock alerts trigger correctly when product quantities fall below the predetermined threshold.
 - Feedback Provided:
 - Improve the UI design of low-stock alerts in the inventory/checkout view.
 - Consider adding filtering options to the product search feature.
 - Priority Revisions:
 1. Inventory reporting features need to be moved higher in priority.
 2. UI refinements moved to a lower priority.
- Testing
 - During this sprint, we implemented several unit tests to verify the functionality of our product's UI and product checkout features. These tests helped ensure that the added features during this sprint function correctly and integrate properly with the checkout process.
 - Unit Tests:
 - Test that adding items updates the total.
 - Test for when the cart is empty, checkout is unavailable.
 - Test that makes sure the cart matches the checkout list.
 - Test that the tax total is computed correctly.
 - Test that UI components render consistently across the system pages (and follow our style guidelines).

Sprint Retrospective:

- Changes in the Scrum Process:
 - Again, our team does not believe we need to make any changes to the Scrum Process yet, in the context of this Sprint.
- Start: Writing unit tests during development rather than after implementation.
- Stop: Delaying UI consistency updates until later stages of development.
- Continue: Frequent commits, clear communications, and consistent task tracking in the SCRUM board.
- Other Planned Adjustments/Concerns:

- The only thing we tried our best to change during this sprint was adding more information to our Sprint Reports. We added multiple new sections to the report to better show what we are doing during each Sprint.
- Team Velocity:
 - Sprint 1: 12 story points
 - Sprint 2: 16 story points
 - Sprint 3: 18 story points
 - Sprint 4: 17 story points
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 - Average Velocity: ~16 story points per sprint

Backlog Grooming

- PBIs/Tasks Added/Removed/Reprioritized
 - Stock reorder suggestions: Give recommendations when inventory levels fall below our defined threshold.
 - Track sales transactions: Record and store completed sales transactions for reporting and auditing.
 - Database updates product levels based on sales transactions: Automatically update inventory quantities when sales occur.
 - Update UI navigation between pages: Improve the navigation between the inventory, checkout, and reporting pages.
- Definition of Ready (DoR)
 - Our DoR: All backlog items/tasks selected for the next sprint meet our defined acceptance criteria, estimated story points, identified dependencies, and can be reasonably completed within one sprint.

Daily Standup #1:

- Duration: 5 Minutes
- Facilitator: Jason Gun

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Tyrese Dickson

1. **What did I do yesterday?**
I wrapped up working on the 'add items' to the database functionality.
2. **What do I plan to do today?**
Today, I planned out how I am going to ensure the frontend UI is consistent between each page and follows our style guidelines.
3. **What are any challenges or impediments that I have experienced?**
None

Timyathus Spikes

1. **What did I do yesterday?**
I wrapped up working on the 'remove items' functionality in the database.
2. **What do I plan to do today?**
Today, I planned out how I was going to implement the 'search' for products via name or SKU if the barcode scanner fails.
3. **What are any challenges or impediments that I have experienced?**
None

Jason Gun

1. **What did I do yesterday?**

I wrapped up working on creating the functional sales table.

2. **What do I plan to do today?**

Today, I planned out how I am going to implement barcode scanning so that items are automatically added to the cart when scanned.

3. **What are any challenges or impediments that I have experienced?**

None

Daniel Foister

1. **What did I do yesterday?**

I wrapped up working on role-based authentication for inventory management and checkout.

2. **What do I plan to do today?**

Today, I planned out how I am going to implement low stock level alerts within the system.

3. **What are any challenges or impediments that I have experienced?**

None

Daily Standup #2:

- Duration: 5 Minutes
- Facilitator: Jason Gun

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Tyrese Dickson

4. **What did I do yesterday?**

Yesterday, I planned out how I was going to ensure the frontend UI is consistent between each page and follows our style guidelines.

5. **What do I plan to do today?**

Today, I am going to begin implementing the frontend UI updates to ensure consistency across all pages based on the style guidelines, which I planned out during the last session.

6. **What are any challenges or impediments that I have experienced?**

None

Timyathus Spikes

4. **What did I do yesterday?**

Yesterday, I planned out how I was going to implement the 'search' for products via name or SKU if the barcode scanner fails.

5. **What do I plan to do today?**

Today, I am going to implement the search functionality that allows users to look up products by name or SKU when the barcode scanner fails, which I planned out during the last session.

6. **What are any challenges or impediments that I have experienced?**

None

Jason Gun

4. **What did I do yesterday?**

Yesterday, I planned out how I was going to implement barcode scanning so that items are automatically added to the cart when scanned.

5. **What do I plan to do today?**

Today, I am going to begin implementing the barcode scanner functionality so

that scanned items are automatically added to the cart, which I planned out during the last session.

6. **What are any challenges or impediments that I have experienced?**

None

Daniel Foister

4. **What did I do yesterday?**

Yesterday, I planned out how I was going to implement low stock level alerts within the system.

5. **What do I plan to do today?**

Today, I am going to implement the low stock alert functionality that notifies users when inventory levels fall below a defined threshold, which I planned out during the last session.

6. **What are any challenges or impediments that I have experienced?**

None

Daily Standup #3:

- Duration: 5 Minutes
- Facilitator: Jason Gun

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Tyrese Dickson

7. **What did I do yesterday?**

I continued working on implementing frontend UI updates to ensure consistency between each page and alignment with our style guidelines.

8. **What do I plan to do today?**

Complete my assigned task and plan for Sprint #5.

9. **What are any challenges or impediments that I have experienced?**

None

Timyathus Spikes

7. **What did I do yesterday?**

I continued working on implementing the product search functionality by name or SKU in case the barcode scanner fails.

8. **What do I plan to do today?**

Complete my assigned task and plan for Sprint #5.

9. **What are any challenges or impediments that I have experienced?**

None

Jason Gun

7. **What did I do yesterday?**

I continued working on implementing the barcode scanning functionality to automatically add items to the cart.

8. **What do I plan to do today?**

Complete my assigned task and plan for Sprint #5.

9. **What are any challenges or impediments that I have experienced?**

None

Daniel Foister

7. **What did I do yesterday?**

I continued working on implementing the low stock level alert functionality within the system.

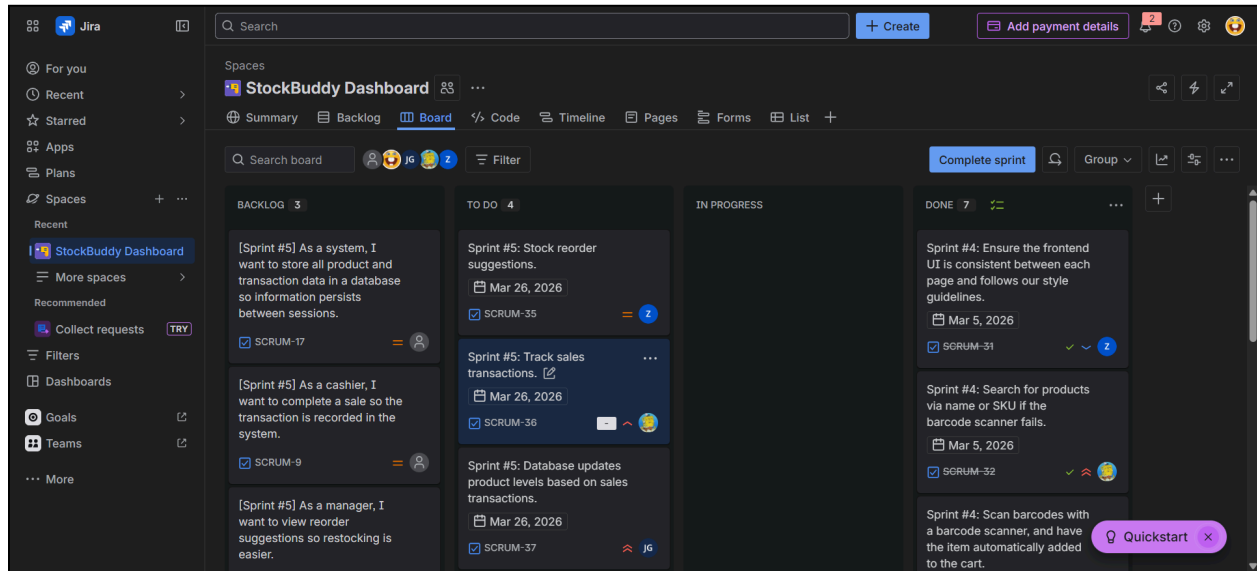
8. **What do I plan to do today?**

Complete my assigned task and plan for Sprint #5.

9. **What are any challenges or impediments that I have experienced?**

None

Exported Copy of Scrum Board:



END.